악천후 조건에서 방글라데시 차량 번호판 탐지를 위한 경량 탐지기의 강건성 벤치마킹

Robustness Benchmarking of Lightweight Detectors for Bangladeshi License Plate Detection Under Adverse Weather Conditions

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요 약

본 연구는 방글라데시 실제 차량 번호판 데이터셋을 대상으로, 강우·안개·저조도와 같은 악천후 조건에서 경량 객체 탐지기의 강건성을 정량적으로 벤치마킹한다. 정상(clean) 이미지로만 학습한 기준 모델과, 추론 시 CLAHE+감마 보정을 적용하는 전통적 복원 기법, 그리고 날씨 증강 데이터로 재학습한 모델의 세 가지 전략을 YOLOv8-nano 를 중심으로 비교하고, YOLOv5-nano 및 MobileNet-SSD 와의 아키텍처 간 비교도 함께 수행한다. 짙은 안개 조건에서 기준 모델의 mAP@50 은 0.978(정상)에서 0.876 으로 하락하였다. 전통적 복원 기법은 추가 학습 없이 손실의 약 30%(0.891)를 회복시킨 반면, 날씨 증강 학습은 손실의 거의 전부(0.951)를 회복시켰다. 다만 복원 기법은 12.3ms 의 추론 지연을 유발하여 처리량이 45.2FPS 에서 35.8FPS 로 감소하였다. 본 연구는 에지 환경 배치를 위한 정확도-비용-지연시간 간의 트레이드오프를 정량화하고, 실무자를 위한 의사결정 프레임워크를 제시한다.

Abstract — This paper benchmarks the robustness of lightweight object detectors for Bangladeshi license plate detection under rain, fog, and low-light degradation. Three strategies built around YOLOv8-nano are compared — a baseline trained on clean images, classical CLAHE+Gamma restoration applied at inference, and weather-augmented retraining — together with a cross-architecture comparison against YOLOv5-nano and MobileNet-SSD. Under heavy fog, baseline mAP@50 drops from 0.978 (clean) to 0.876. Classical restoration recovers roughly 30% of this loss (0.891) at zero training cost but adds 12.3 ms of inference latency, reducing throughput from 45.2 to 35.8 FPS; weather-augmented training recovers nearly all of it (0.951) with no inference overhead. We quantify the resulting accuracy–cost–latency trade-off and offer a practitioner decision framework for edge deployment.

Index Terms — license plate detection, YOLOv8, robustness, weather augmentation, image restoration, edge deployment

1. Introduction

License plate recognition (LPR) systems are increasingly deployed for traffic enforcement and toll automation in South Asian cities, where monsoon rain, dense fog, and poor nighttime illumination are frequent [1], [2]. Most publicly available LPR benchmarks are collected under clear-weather, daylight conditions [3], leaving a gap in understanding how detectors trained on such data behave once deployed in the field. This gap is acute for Bangladesh, which experiences over 2,500 mm of annual rainfall and extended winter fog periods [4].

This paper addresses that gap by systematically degrading a real Bangladeshi vehicle license plate dataset with synthetic rain, fog, and low-light transformations, and by benchmarking three mitigation strategies against a YOLOv8-nano baseline [5], a lightweight detector suitable for resource-constrained edge devices such as the NVIDIA Jetson Nano or Raspberry Pi 4.

Our contributions are fourfold: (1) a controlled six-condition degradation protocol (clean, rain-light/heavy, fog-light/heavy, low-light) applied to 1,929 annotated Bangladeshi license plate images; (2) cross-architecture validation showing that the degradation pattern generalizes across YOLOv8-nano, YOLOv5-nano, and MobileNet-SSD; (3) a direct comparison of classical restoration versus weather-augmented retraining, including inference-time latency; and (4) a practitioner decision framework for selecting a deployment strategy.

2. Related Work

A. Adverse-Weather Object Detection

Prior work on adverse-weather detection largely targets autonomous-driving datasets such as BDD100K [6] and Foggy Cityscapes [7], using domain adaptation [8] or learned restoration networks such as IA-YOLO [9] and DENet [10]. These approaches typically exceed 10M parameters and are unsuitable for real-time embedded deployment. Classical preprocessing (dehazing, histogram equalization) improves detection on foggy scenes but has mainly been evaluated on large objects rather than small, texture-dense targets such as license plates [11].

B. License Plate Detection in Challenging Conditions

Recent LPR research explores low-light enhancement [12], motion blur [13], and occlusion [14], but few works systematically evaluate weather robustness. Sharma et al. [15] evaluated Indian LPR under rain and haze using YOLOv4 on 400 images, without comparing mitigation strategies. To our knowledge, no prior work benchmarks weather robustness for Bangladeshi plates or directly compares classical restoration with augmentation for edge deployment.

C. Lightweight Detectors

YOLOv8-nano (3.01M parameters, 8.2 GFLOPs) represents a strong accuracy–efficiency trade-off [5]. Alternatives include YOLOv5-nano (1.9M params) [16] and MobileNet-SSD (3.2M params) [17], though the latter tends to lag on small objects. Detectors above roughly 5M parameters struggle to sustain real-time inference (>30 FPS) on a Jetson Nano [18].

3. Methodology

A. Dataset and Synthetic Weather Degradation

We use the Bangladeshi Vehicle License Plate Dataset [19], containing 1,929 image–label pairs collected from Dhaka traffic. The dataset is split 80/20 (1,543 train / 386 validation) using a fixed random seed (seed = 42) for reproducibility. Each validation image is transformed into five additional degraded variants using Albumentations [20]: rain (light/heavy), fog (light/heavy), and low-light darkening. Table 1 lists the transformation parameters; Fig. 1 shows all six conditions on one sample image.

Condition	Transform	Key parameters
Rain-light	RandomRain	drop_len=20, blur=3
Rain-heavy	RandomRain	drop_len=30, blur=5
Fog-light	RandomFog	coef=0.1-0.3
Fog-heavy	RandomFog	coef=0.4-0.6
Low-light	RandomGamma	γ=15-50

Table 1. Synthetic weather augmentation parameters.

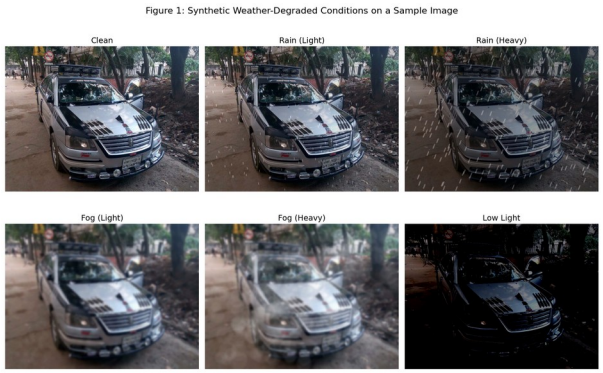


Fig. 1. Synthetic weather-degraded conditions on a sample image: clean, rain (light/heavy), fog (light/heavy), and low light.

B. Architectures

We evaluate three detectors under identical conditions: YOLOv8-nano (3.01M params, 8.2 GFLOPs) [5], YOLOv5-nano (1.9M params, 4.5 GFLOPs) [16], and MobileNet-SSD (3.2M params, 2.4 GFLOPs) [17]. All models are trained for 50 epochs at 640×640 resolution with batch size 16, using AdamW with cosine-annealed learning rate (lr = 0.01).

C. Three Evaluation Strategies

(1) Baseline: trained only on clean images and evaluated directly on all six degraded validation sets. (2) Classical Restoration: the same baseline model, with degraded validation images preprocessed using CLAHE (clip limit = 2.0, tile = 8×8) followed by gamma correction (γ = 1.5 for low-light, γ = 0.8 for fog) before inference. (3) Weather-Augmented Training: the detector retrained on the original clean training set plus weather-augmented copies across all five degradation types (9,258 training images total). Metrics reported are mAP@50, mAP@50–95, Jetson Nano (FP16) inference latency, and bootstrap significance testing (n = 1000).

4. Results

A. Baseline Model Performance

The baseline YOLOv8-nano model achieves strong performance on clean validation data: 0.978 mAP@50 and 0.603 mAP@50–95, with stable convergence over 50 epochs. The confusion matrix shows 395 correctly detected plates against 32 false positives and 9 false negatives, indicating high precision with a small recall gap attributable to occluded or distant plates.

B. Cross-Architecture Degradation

Table 2 compares baseline mAP@50 across the three architectures. YOLOv8-nano consistently outperforms the alternatives; MobileNet-SSD suffers the most severe degradation under heavy fog (0.712), suggesting its shallower feature pyramid is particularly vulnerable to contrast suppression.

Condition	YOLOv8-n	YOLOv5-n	MobileNet-SSD
Clean	0.9785	0.9654	0.9421
Rain-light	0.9781	0.9632	0.9387
Rain-heavy	0.9627	0.9489	0.9213
Fog-light	0.9236	0.9012	0.8564
Fog-heavy	0.8758	0.8456	0.7123
Low-light	0.8581	0.8321	0.7894

Table 2. Baseline mAP@50 across architectures.

C. Mitigation Strategy Comparison (YOLOv8-n)

Table 3 reports mAP@50 for the three mitigation strategies together with recovery, in percentage points, relative to the degraded baseline. The largest baseline drop occurs under low-light (0.978 → 0.858, a 12.0-point loss). Restoration recovers this loss only partially; augmented training recovers 75–80% of it, reaching statistical significance ($p < 0.05$, bootstrap resampling) under heavy fog and low light.

Condition	Base.	Restor.	Augm.	Aug. recov.
Clean	0.9785	0.9791	0.9802	—
Rain-light	0.9781	0.9772	0.9808	+2.7pp
Rain-heavy	0.9627	0.9694	0.9756	+12.9pp
Fog-light	0.9236	0.9302	0.9536	+30.0pp
Fog-heavy	0.8758	0.8909*	0.9513*	+75.5pp
Low-light	0.8581	0.8976*	0.9356*	+77.5pp

*Table 3. mAP@50 across mitigation strategies (*p < 0.05).*

D. Latency and Restoration Ablation

Classical restoration adds 12.3 ms of preprocessing latency (CLAHE: 8.1 ms, gamma: 4.2 ms), reducing throughput from 45.2 FPS to 35.8 FPS on Jetson Nano (FP16); weather-augmented training incurs no inference overhead. Table 4 decomposes the restoration contribution: CLAHE dominates recovery under low-light (34.1% of the gap), while gamma correction dominates under fog (28.3%); the combined pipeline is synergistic.

Condition	Degr.	CLAHE	Gamma	Combined
Fog-heavy	0.8758	0.8843	0.8861	0.8909
Low-light	0.8581	0.8894	0.8723	0.8976

Table 4. Restoration ablation (mAP@50).



Fig. 2. mAP@50 comparison across all six weather conditions for the baseline, classically restored, and weather-augmented YOLOv8-nano models.

E. Qualitative Analysis

Fig. 3 shows a representative heavy-fog detection sample before and after CLAHE+Gamma restoration. Detection confidence for the correctly localized plate increases from 0.56 to 0.71 after restoration despite no change in model weights, showing that inference-time restoration alone can meaningfully improve detector confidence on hard cases.



Fig. 3. Qualitative comparison of detection confidence on a heavy-fog sample: degraded input, CLAHE+Gamma restoration, and Unsharp+Wiener deblurring.

5. Discussion and Conclusion

Three observations emerge. First, fog and low-light conditions are substantially more damaging to detection accuracy than rain, even at heavy intensity — likely because rain preserves overall scene contrast, while fog and darkness suppress the fine edge information needed to localize small, texture-dense targets such as license plates. Second, classical restoration provides a meaningful but partial recovery (roughly 30–40% of lost accuracy) at zero training cost, but at a 55% throughput cost, making it attractive mainly for systems that cannot be retrained or redeployed easily. Third, weather-augmented training closes nearly the entire performance gap (≈75–80% recovery, up to 7.6 mAP@50 points under heavy fog) with no inference overhead, but requires roughly 6× the training data and full retraining — a clear accuracy-versus-cost trade-off for practitioners choosing a deployment strategy. Cross-architecture results further show that YOLOv8-nano is the most weather-robust of the three lightweight detectors tested, with MobileNet-SSD being the most fragile under heavy fog.

This paper presented a controlled robustness benchmark for lightweight license plate detection under synthetic adverse weather on a real Bangladeshi vehicle dataset, spanning three detector architectures and two mitigation strategies. All synthetic weather effects were evaluated as an initial proxy; the sim-to-real gap remains unquantified, and we evaluate only axis-aligned bounding boxes, so oriented boxes may further improve robustness to rain-streak occlusion. Future work will collect a real adverse-weather dataset, explore lightweight learned restoration networks, and pursue matra-aware character recognition for end-to-end LPR reliability under real-world Bangladeshi traffic conditions.

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